# Unit 1

# Measurement, Limits & Fits, Gauges

(Handwritten Notes)

## Syllabus Topics Covered

1. Concept of measurement, feedback control, generalized measurement system
2. Units and standards
3. Measuring instrument characteristics: sensitivity, accuracy, precision, response
4. Errors, correction and calibration
5. Limits, fits and tolerances; hole & shaft basis; interchangeability
6. Limit gauges, Taylor's principle, gauge design
7. Linear measurement: vernier, micrometer, slip gauges, tool maker's microscope
8. Comparators: mechanical, pneumatic, electrical
9. Angular measurement: sine bar, bevel protractor, angle dekkor, rollers & spheres

## 1. Concept of Measurement

**Metrology** is the science of measurement. **Measurement** is the process of **comparing an unknown quantity with a known standard** of the same kind, to find how many times the standard is contained in the unknown quantity.

Result of a measurement

**Measurement = Number x Unit** (e.g., length = 25.4 x mm)

### Objectives of metrology

- To evaluate **newly developed products** and judge their performance, and to determine the **process capabilities** .
- To determine the **measuring instrument capabilities** and ensure they are adequate for the application.

- To **minimize inspection cost** and the cost of rejection; to maintain **interchangeability** of parts; to **standardize** methods and the gauging practice.
- To obtain the **quality** level of the manufacturing process through **feedback** , and to establish the **traceability** of measurements to national standards.

## Methods of measurement

| Method | Description | Example |
|---|---|---|
| Direct | value obtained directly by comparing with a standard | steel rule, vernier caliper |
| Indirect | value computed from other measured quantities | area = l x b; sine bar angle |
| Comparative | compared with a known standard; only deviation shown | dial indicator + slip gauges |
| Absolute / fundamental | based on measurement of base quantities | dead-weight pressure gauge |

## Generalized measurement system

Any measuring system can be broken into **functional elements** (stages):

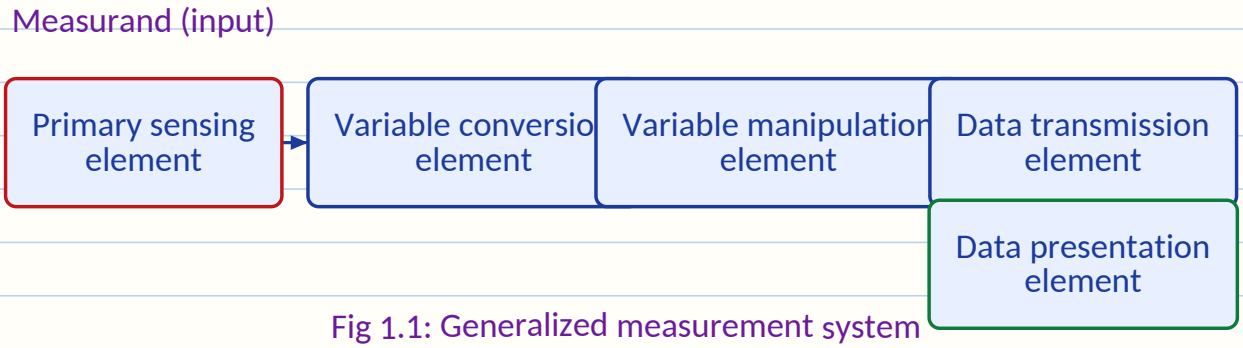



Fig 1.1: Generalized measurement system

1. **Primary sensing element** - senses the quantity ( **measurand** ) and gives output (e.g., bellows, bimetallic strip, Bourdon tube).
2. **Variable conversion element** - converts the output to a more convenient physical variable (e.g., strain gauge, LVDT).
3. **Variable manipulation element** - amplifies / modifies the signal (amplifier, gear train, lever).
4. **Data transmission element** - carries the signal from one place to another (cables, telemetry, linkages).
5. **Data presentation element** - displays the result (dial, scale, recorder, printer, CRO, digital display).

## Concept of feedback control system

A **control system** keeps an output at a desired value by **feeding back** the measured output and comparing it with the reference (set point). Measurement is the **feedback element** of the loop.

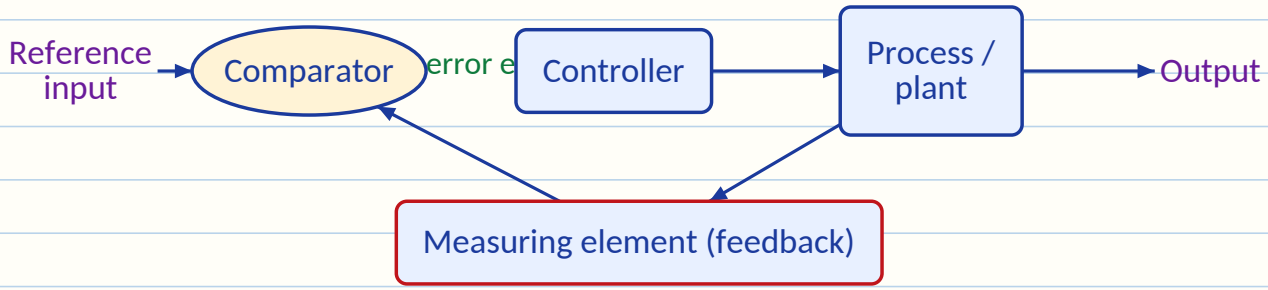



Fig 1.2: Feedback control system (closed loop)

- **Error = Reference - Feedback** . Controller acts to reduce the error. Examples: thermostat, speed governor, CNC machine position feedback by encoder.
- **Open-loop** system has no feedback (washing machine timer) - cannot correct disturbances.

## 2. Units and Standards

A **unit** is a definite magnitude of a physical quantity used for comparison. A **standard** is the physical embodiment of a unit. The **SI** (Systeme International) is the universally accepted system.

| Base quantity | SI unit | Symbol |
|---|---|---|
| Length | metre | m |
| Mass | kilogram | kg |
| Time | second | s |
| Electric current | ampere | A |
| Temperature | kelvin | K |
| Luminous intensity | candela | cd |
| Amount of substance | mole | mol |

- **Supplementary / derived** : plane angle - radian; solid angle - steradian; force - newton (N); pressure - pascal (Pa); energy - joule (J); power - watt (W).
- **Metre** (current definition): the length of the path travelled by light in vacuum during **1/299 792 458 s** . Earlier: international prototype (Pt-Ir bar, 1889) and

1,650,763.73 wavelengths of Kr-86 orange-red line (1960).

## Types of standards

1. **Line standard** : length defined as the distance between **two engraved lines** on a bar (Imperial standard yard, International prototype metre).
2. **End standard** : length defined between **two flat parallel faces** (slip gauges, end bars, micrometer anvils).
3. **Wavelength standard** : length defined in terms of wavelength of light (krypton-86, laser) - **not** subject to wear or damage; reproducible anywhere.

| Line standard | End standard |
|---|---|
| Distance between two lines | Distance between two flat faces |
| Quick and easy to use; parallax error | No parallax; wringing needed; better accuracy |
| Not so accurate (+-0.2 mm to 0.1 mm) | Highly accurate (to 0.0005 mm) |
| Subject to scale wear / marking errors | Faces wear with use; temperature effects |
| Steel rule, metre bar | Slip gauge, vernier jaws, micrometer |

- **Primary** : kept at the national / international level, never used directly. **Secondary** : calibrated against primary, used to calibrate others. **Working** : used daily in workshops (slip gauges grade 2, gauges).

## 3. Measuring Instrument Characteristics

| Term | Meaning |
|---|---|
| **Sensitivity** | ratio of change in output (pointer movement) to change in the measured input: S = dO/dI. Higher magnification -> higher sensitivity. Also the smallest input change that gives a perceptible output. |
| **Readability** | ease with which the readings can be taken, depends on the scale spacing, pointer thickness and parallax. Vernier / digital is easier to read. |
| **Range** | difference between the maximum and minimum values the instrument can measure (e.g., 0-25 mm micrometer). **Span** = max - min. **Least count (resolution)** = smallest reading possible. |
| **Accuracy** | closeness of the measured value to the **true value** . Expressed as +-% of full scale / reading. |

| | |
|---|---|
| **Precision** | closeness of **repeated readings** to each other - **repeatability** , not related to the true value. |
| **Repeatability** | ability of the instrument to give the same reading when the same quantity is measured repeatedly, under same conditions, short time interval. |
| **Hysteresis** | difference in reading for the same input when approached from above and below (friction, backlash). |
| **Magnification** | ratio of the length of the pointer (scale) movement to the change in size of the work. Linear: sensitivity = magnification. |

## Static and dynamic response

- **Static characteristics** : those for measurements of **constant or slowly varying** quantities - accuracy, sensitivity, hysteresis, range, resolution, linearity, repeatability.
- **Dynamic characteristics** : how the instrument responds to **rapidly changing** input:
  - **Speed of response** - how quickly the instrument reacts to a change in the measured quantity.
  - **Measuring lag** - delay in response (retardation type: begins at once; time-delay type: begins after a dead time).
  - **Fidelity** - ability to reproduce the input faithfully, without dynamic error.
  - **Dynamic error** - difference between the true value of the time-varying quantity and the value indicated (assuming zero static error).

# 4. Errors, Correction and Calibration

### Error

**Error = Measured value - True value** . **Correction = -(Error)** , i.e. True value = Measured value + Correction.

- Absolute error = |measured - true|; Relative (percentage) error = (error / true value) x 100.

## Types of errors

1. **Systematic (fixed) errors** : constant in magnitude and sign, **predictable** and can be **eliminated** by correction / calibration. Causes: faulty calibration, zero error, wear of instrument, temperature effect, **loading** errors, **Abbe's** error.

2. **Random errors** : vary in magnitude and sign **unpredictably** - caused by small changes in temperature, humidity, vibration, operator judgement. Cannot be eliminated; treated **statistically** (mean, standard deviation); reduced by averaging many readings.
3. **Gross errors** : human mistakes - wrong reading, wrong recording, misuse of instrument.

| Systematic errors | Random errors |
|---|---|
| Same magnitude/sign each time | Vary in magnitude and sign |
| Can be determined and corrected | Cannot be corrected; estimated statistically |
| Affect accuracy | Affect precision |
| Eg. zero error, wrong calibration | Eg. thermal fluctuations, vibration |

## Calibration

**Calibration** is the process of comparing an instrument against a **standard of higher accuracy** under specified conditions, to determine its error and apply corrections or adjust it.

- Steps: select the standard -> apply known inputs over the range (up and down) -> record the instrument readings -> prepare the **calibration curve / table** -> compute errors, hysteresis -> adjust, certify and label the next calibration due date.
- **Traceability** : unbroken chain of calibrations to the national / international standard (NPL India). Calibrate at intervals; after any shock or repair.

# 5. Limits, Fits and Tolerances

No two parts can be made **exactly** the same size - some variation is unavoidable. A **tolerance** is the permissible variation in a dimension. The designer specifies **limits** so that parts function and **interchange** .

## Terminology (hole and shaft)

| Term | Definition |
|---|---|
| Basic size (nominal) | size from which limits are derived; theoretical size, e.g. 40 mm |
| Actual size | size obtained by measurement |
| Limits of size | maximum ( **upper limit** ) and minimum ( **lower limit** ) permissible sizes |
| Tolerance | difference between upper & lower limits = **UL - LL** |

| | |
|---|---|
| Deviation | algebraic difference between a size and the basic size (upper / lower deviation) |
| Zero line | line representing the basic size; deviations +ve above, -ve below |
| Fundamental deviation | deviation nearest to the zero line (decides the position of tolerance zone) |
| MMC / LMC | maximum material condition (smallest hole, largest shaft) / least material condition |
| Allowance | intentional difference between the MMC of mating parts = min. clearance (+) or max. interference (-) |

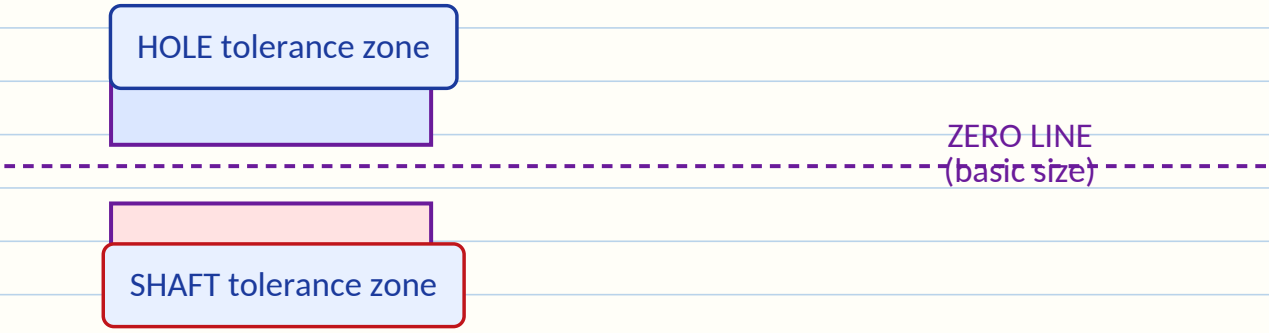



Fig 1.4: Tolerance zones of hole and shaft relative to the zero line

## Types of tolerance

- **Unilateral** : tolerance in **one direction** only from basic size: 25 +0.02/-0.00.
  **Bilateral** : on both sides: 25 +-0.02 (equal) or +0.03/-0.01 (unequal).

## Fits

**Fit** = relationship between the sizes of mating hole and shaft (before assembly).

1. **Clearance fit** : shaft always **smaller** than hole -> always a clearance. Free movement (bearings, sliding fits). E.g. H7/g6 (running), H8/f7.
2. **Interference fit** : shaft always **larger** than hole -> press / shrink fit; permanent assembly (bush in housing). E.g. H7/p6, H7/s6.
3. **Transition fit** : may be clearance or interference depending on actual sizes (location fit: keys, gear on shaft). E.g. H7/k6, H7/n6.

| Fit | Clearance (max / min) | Condition |
|---|---|---|
| Clearance | Max C = HML hole - LML shaft; Min C = LML hole - HML shaft | Hole LL >= Shaft UL |

| | | |
|---|---|---|
| Interference | Max I = HML shaft - LML hole; Min I = LML shaft - HML hole | Shaft LL >= Hole UL |
| Transition | Max C and Max I both exist | Tolerance zones overlap |

## Hole basis and shaft basis systems

### Hole-basis system

Hole size is kept **constant** (fundamental deviation = 0, letter **H** : lower deviation zero) and the **shaft size varied** to get the different fits. **Preferred** because holes are produced by **fixed-size tools** (drills, reamers, broaches) and checked with plug gauges, while shafts are easily turned to any size.

### Shaft-basis system

Shaft size is kept **constant** (letter **h** : upper deviation zero) and the hole size is varied. Used when **one shaft** carries many parts with different fits (cold-drawn bar, line shafting with pulleys, couplings).

## ISO system of limits and fits

- **Tolerance grades** : **IT01, IT0, IT1 ... IT16** (20 grades). Lower number = finer. Gauges IT1-IT5, fits IT5-IT11, rough IT12-IT16. IT = i x multiplier where i (standard tolerance unit) = 0.45 D^1/3 + 0.001 D (microns, D = geometric mean of size range in mm).
- **Fundamental deviation letters** : holes **A...ZC** (capital), shafts **a...zc** (small). H/h = zero deviation. 28 positions.
- Designation: **40 H7/g6** -> basic size 40 mm, hole tolerance grade 7 with H position, shaft grade 6 with g position.

## Interchangeability

- **Interchangeability** - parts made to the specified limits can be assembled or replaced **without selection, fitting or adjustment** . Basis of mass production, spares availability and low cost.
- **Universal (full) interchangeability** : any component fits any mate (screws, bolts, bearings). **Selective interchangeability / selective assembly** : parts are sorted into groups by size (piston-cylinder) - for very tight tolerances at low cost.

# 6. Limit Gauges and Gauge Design

A **limit gauge** checks whether a dimension lies **within its tolerance limits** (no numerical value is read). It has two members: **GO** end (checks the **MMC** ; must pass) and **NO GO / NOT GO** end (checks the **LMC** ; must not pass). Fast, cheap, usable by unskilled workers.

## Types of gauges

| Gauge | Use | Notes |
|---|---|---|
| Plug gauge | checks **holes** (diameter) | double-ended; GO plug long, NO GO plug short (shorter & chamfered); handle taper |
| Ring gauge | checks **shaft** diameter / threads | GO ring and NO GO ring (with groove on rim to identify) |
| Snap (caliper) gauge | checks **shafts** , widths, thickness | double-ended or progressive; adjustable snap gauges |
| Gap gauge | checks lengths between faces / thickness | fixed gap for large work |
| Taper gauges | taper plug & ring gauges | check taper & diameter of Morse tapers (contact blue) |

## Taylor's principle

### Taylor's principle

- The **GO gauge** should check **all features** of the dimension **at the same time** (full form, e.g. a full-length cylindrical plug checks diameter, roundness, straightness) - the GO gauge must be **full form** .
- The **NO GO gauge** should check **only one feature at a time** (separate NO GO for each dimension; point contact) - so a defect in one feature is not masked by another.

- Reason: if a NO GO plug were full form, it might not enter because of an **ovality** error even when the hole size is over the upper limit - then a bad hole would be accepted. A short NO GO plug can enter at a large diameter and reject.

## Design of GO and NO GO gauges

- **Gauge tolerance** : gauges themselves can't be made perfectly. Usually **10% of work tolerance** (IS 919: gauge tolerance depends on the grade). Tolerance zone

of gauges is **inside** the work tolerance zone so that no bad part is accepted.

- **Wear allowance** : GO gauges wear since they are used at every check - provide extra material = **5% of the work tolerance** (located between the work limit and the gauge tolerance zone). NO GO gauge needs **no** wear allowance (seldom enters).

Design example (Plug gauge for hole 25 +0.021 / 0)

Work tolerance = 0.021 mm; Gauge tolerance = 10% x 0.021 = **0.0021** ; Wear allowance = 5% x 0.021 = **0.00105** .

- GO plug: lower limit = 25.000 + 0.00105 = **25.00105** ; upper limit = 25.00105 + 0.0021 = **25.00315** (wear allowance applied inside the work tolerance).
- NO GO plug: upper limit = 25.021 (HML); lower limit = 25.021 - 0.0021 = **25.0189** .

- **Snap gauge for shaft** (e.g. 25 f8): GO snap = HML of shaft - wear allowance (and - gauge tol.), NO GO snap = LML + gauge tolerance (inside). Same logic mirrored for the shaft.

# 7. Linear Measurement

## 7.1 Vernier instruments

Based on the **vernier principle** : two scales of slightly different divisions - the main scale (MS) and sliding vernier scale (VS) - enable reading to a fraction of the smallest main division.

Least count of vernier

**LC = 1 MSD - 1 VSD = (value of 1 main-scale division) / (number of vernier divisions)**

- Example: 1 MSD = 1 mm; 50 VSD coincide with 49 MSD -> LC = 1/50 = **0.02 mm** .
- **Reading = MSR + (VSC x LC)** , where MSR = main-scale reading, VSC = coinciding vernier division.

- Types: **vernier caliper** (outside, inside, depth), **vernier height gauge** (with scriber; for marking & height on surface plate), **vernier depth gauge** , **gear-tooth vernier** (chordal thickness).
- Parts of caliper: fixed jaw with beam, sliding jaw, locking screw, fine adjustment, depth rod; **inside jaws** (knife edges).

- **Errors** : zero error, parallax, **Abbe's-principle violation** (jaws away from scale), measuring force, wear and jaw non-parallelism.

Example: MSR = 24 mm, 17th vernier division coincides, LC = 0.02 -> Reading = 24 + 17 x 0.02 = **24.34 mm** (zero error corrected).

## 7.2 Micrometers

Works on the **screw-nut principle** : when the screw is rotated one revolution through a fixed nut, it advances axially by one pitch. Pitch = 0.5 mm, thimble has 50 divisions -> **LC = 0.5/50 = 0.01 mm** .

- **Parts** : **frame (C-shaped)** , anvil, spindle, sleeve (barrel with datum line), thimble (circular scale), **ratchet stop** (constant measuring force), locknut.

**Reading = Sleeve reading + (thimble division x 0.01)** .

## 7.3 Slip gauges (gauge blocks)

**End standards** : rectangular blocks of **hardened steel / ceramic / carbide** , with two opposite faces **lapped flat and parallel** to a very high degree - used to set, calibrate other instruments, and for direct precise measurement. Invented by **C.E. Johansson** .

- **Wringing** : when two clean lapped faces are slid together with slight pressure and twist, they **adhere** by **molecular attraction + thin oil film** (force ~ 300 N). Size of the stack = sum of block sizes (wringing film ~ 0.00001 mm).
- **Grades** (BIS / IS 2984): **Grade 00** (reference / master, +-0.05 micron), **Grade 0** (inspection / calibration), **Grade 1** (workshop / toolroom), **Grade 2** (workshop). Standard sets: M87 (87 pieces), M112, M45, M33.
- Sizes: 1.001-1.009 (9 pcs, 0.001 step), 1.01-1.49 (49 pcs, 0.01 step), 0.5-24.5 (49 pcs, 0.5 step), 25-100 (4 pcs, 25 step) - M87 set.

Building a dimension: 41.125 mm (M87)

Choose from the **last decimal** to eliminate digits one by one:

- 41.125 - 1.005 = 40.12 -> - 1.02 = 39.1 -> - 1.1 = 38 -> - 38 (take 38 mm).
- **Blocks: 1.005 + 1.02 + 1.1 + 38 = 41.125 mm** (use as few blocks as possible).

## 7.4 Tool maker's microscope

A **versatile optical measuring instrument** for measuring **dimensions, angles, threads, profile and form** of small, delicate parts (cutting tool profiles, gauges, screw threads, templates) without contact.

- **Construction** : base with a **work table** (XY micrometer heads, 25 mm travel, least count 0.01 mm), glass plate, vertical **column** with microscope tube,

eyepiece with a cross-line reticle (rotating with angle scale 0-360 degrees, vernier 1 minute), illumination (lamp, filters).

## 8. Comparators

A comparator is a relative (comparative) instrument which compares the unknown dimension with a standard (slip gauge stack / master) and shows only the deviation on an amplified scale. It is not used for direct measurement .

Principle

Standard is set on the table -> comparator zeroed -> work replaces the standard -> pointer shows deviation (+/-).

- Uses : inspection of large-lot production , checking of parallelism, roundness, flatness, concentricity , calibrating gauges.

### A. Mechanical comparators

- Dial indicator : plunger with rack moves a pinion -> gear train -> pointer. Magnification ~ 100-1000. Range 0-10 mm, LC 0.01 mm (0.001 mm). Stand with base and column.

### B. Pneumatic comparators

Works on the change in air flow / back pressure as the gap between a nozzle (jet) and the work surface changes. Compressed air at constant pressure supplied through control orifice to a measuring jet .

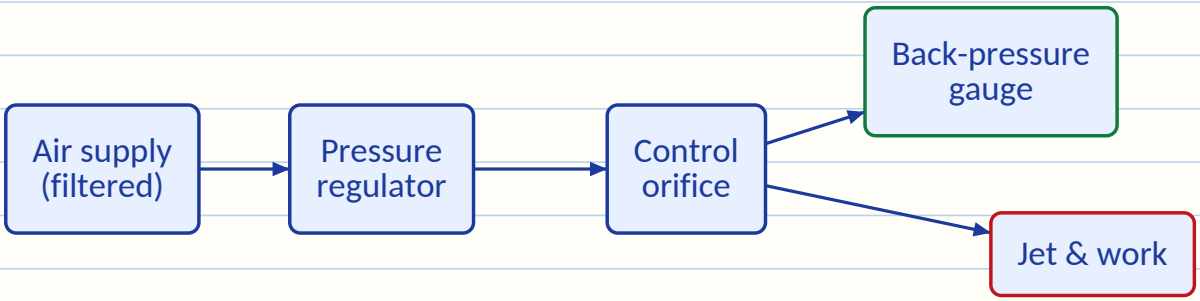



Fig 1.5: Solex back-pressure pneumatic comparator

- Back-pressure type (Solex) : back pressure P between orifice and jet changes with the gap; read by a manometer / dial. Linear between P1 and P2 region -> magnification up to 40,000 .
- Flow type : the flow rate through the gap is measured with a tapered glass tube (rotameter) - float height shows the gap.

### C. Electrical and electronic comparators

- **LVDT** (Linear Variable Differential Transformer): plunger carries a **ferromagnetic core** between a primary and two secondary coils; displacement unbalances the secondaries -> output voltage proportional to displacement. Magnification $10^4$-$10^5$; resolution 0.1 micron; supply AC.

## 9. Angular Measurement

Angles are measured in **degrees, minutes and seconds** (1 degree = 60', 1' = 60'') or **radians** . Instruments: **protractors, sine bar, angle gauges, clinometer, autocollimator, angle dekkor** .

### 9.1 Sine bar

A **precision** hardened and ground steel bar with **two cylindrical rollers** of equal diameter fixed at a **known centre distance L** (100, 200, 300 mm), the rollers' axes parallel to the bar's top face. Used to measure and set angles precisely using **trigonometry** and slip gauges.

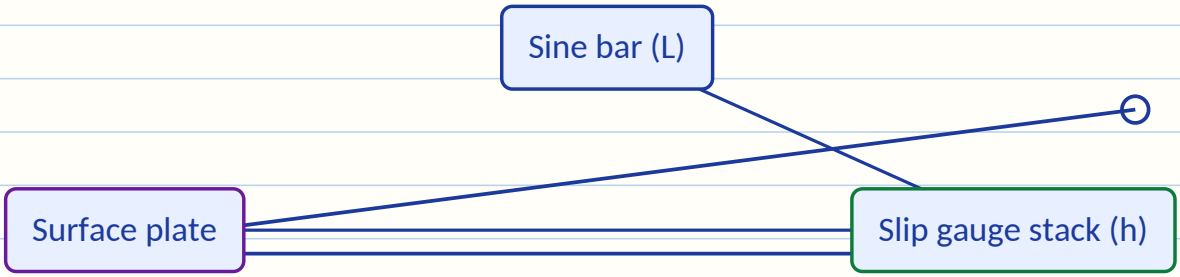



Fig 1.6: Sine bar - sin θ = h / L

Formula

**sin θ = h / L** -> h = L sin θ

- L = distance between roller centres; h = height of slip-gauge stack under one roller.

- **Example** : set a 200 mm sine bar to 30 degrees -> h = 200 x sin 30 = **100 mm** . For 25 degrees: h = 200 x 0.4226 = 84.52 mm.
- **Limitations** : **not suitable for angles > 45 degrees** (error in sine amplifies; as θ -> 90 degree, a small error in h gives a large angle error); **cannot be used for ... large parts** ; accuracy depends on slip gauges and roller diameters; **sensitivity decreases** at larger angles. Types: **sine centre** , **compound sine plate** , sine table, **sine vice** .

## 9.2 Bevel protractor (vernier)

- A **vernier bevel protractor** measures angles to **5 minutes (5')** . Parts: **stock (base)** , **blade** (sliding, adjustable), **dial (circular, 0-90-0 / 360 deg)** , **vernier scale** , lock nut, acute angle attachment.
- **Least count** : 12 vernier divisions = 23 main divisions (1 MSD = 1 degree). 1 VSD = 23/12 degree, so LC = 2 x 1 - 23/12 = 1/12 degree = **5 minutes** .
- Reading = Dial reading + (vernier coinciding division x 5'). Used for angle of **tool cutting edges, V-blocks, dovetails** , bevel gears.

## 9.3 Angle dekkor

An **optical, comparator-type** instrument ( **autocollimator + angle gauges** ) for measuring **small angular differences** ; makes use of **autocollimation** with an **illuminated scale** and a fixed **graticule** .

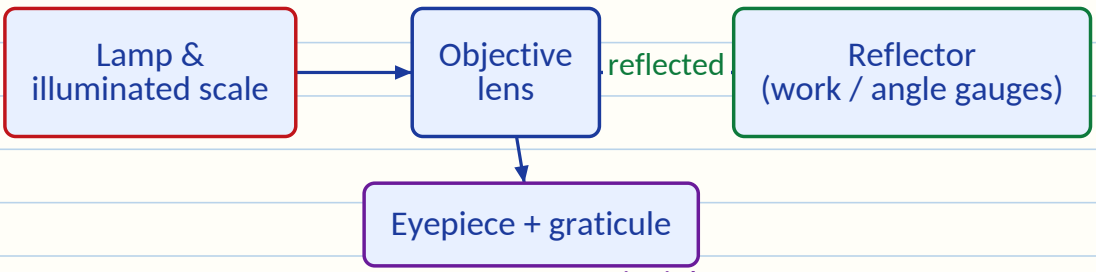



Fig 1.7: Optical principle of angle dekkor

- **Working** : Set the **slip-gauge combination** (angle gauges wrung) equal to the **nominal angle** on the surface plate - **reflector at that angle** ; read scale. Replace the angle gauge by the **work** ; the image shifts: **deviation read on the scale** (1 division = 1 minute; range ~ 1 degree). Image of the **illuminated scale** coincides with the graticule when the reflector is perpendicular to the optical axis.

## 9.5 Rollers and spheres for tapers

**Taper measurement using precision rollers and spheres** , **slip gauges** and a micrometer / height gauge - used for **taper plug gauges, ring gauges, dovetails, cones** .

Using two balls (spheres) in an internal taper (female)

Let R and r = radii of large and small spheres; H = depth difference measured to the tops of the balls (large above small); L = distance between ball centres = **H - R + r** .

**sin (α/2) = (R - r) / L** , where α = included angle of the taper.

Using rollers for a male taper (e.g. taper plug)

Two equal rollers (diameter d) placed on a plane at the large and small ends; measure the distance across them M1 and M2 over rollers at height difference H apart:

**tan (α/2) = (M1 - M2) / 2H** (rollers of same diameter placed at sections); then taper angle = α.

- **Example** : roller measurement: M1 = 68.4, M2 = 60.2 mm, H = 50 mm -> tan (α/2) = (68.4 - 60.2) / (2 x 50) = 0.082 -> α/2 = 4.69 degree; **α = 9.38 degrees** .

## Quick Revision / Important Questions

1. Generalized measurement system; accuracy vs precision; errors and calibration.
2. Limits, fits, tolerances; hole vs shaft basis; Taylor's principle; design of GO / NO GO gauges.
3. Vernier caliper, micrometer, slip gauges (build 41.125 mm), tool maker's microscope.
4. Comparators (mechanical, pneumatic, electrical); sine bar, angle dekkor, bevel protractor; tapers with rollers & spheres.